MILITARY SPECIFICATION

CRACKERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for crackers.

1.2 Classification. The crackers shall be of the following classes.

Class 1	-	Rectangular
Class 2	-	Square
Style A	-	Four crackers (each approximately two inches square, scored but not separated)
Style B	-	One cracker (approximately four by four inches, partially scored)
Class 3	-	Round

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-C-110	-	Cellophane (Coated and Noncoated) Regenerated Cellulose Film
L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8920

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QQ-A-1876 - Aluminum Foil
PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-C-843 - Cushioning Material, Cellulosic
PPP-C-1797 - Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam

STANDARDS

FEDERAL

FED-STD-101 - Preservation, Packaging and Packing Materials: Test Procedures
FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1105 - Sanitary Standards for Bakeries

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 N. Nineteenth Street, Arlington, VA 22209.)

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AMERICAN OIL CHEMISTS SOCIETY

Official and Tentative Methods of the American Oil Chemists Society

(Application for copies should be addressed to the American Oil Chemists Society, 508 South Sixth Street, Champaign, IL 61820.)

AMERICAN PHARMACEUTICAL ASSOCIATON

The National Formulary

(Application for copies should be addressed to the American Pharmaceutical Association, 2215 Constitution Ave., N.W., Washington, DC 20037.)

ASSOCIATION OF VITAMIN CHEMISTS

Methods of Vitamin Assay

(Application for copies should be addressed to the Interscience Publishers, a division of John Wiley & Sons, 605 Third Ave., New York, NY 10016.)

NATIONAL RESEARCH COUNCIL

Food Chemicals Codex

(Application for copies should be addressed to the Office of Director, National Academy of Science, National Research Council, 2101 Constitution Avenue, Washington, DC 20418.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Suite 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 882 - Tensile Properties of Thin Plastic Sheeting
D 1238 - Flow Rates of Thermoplastics by Extrusion Plastometer
D 1505 - Density of Plastics by Density Gradient Technique

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.4 and 6.4).

3.2 Material. All materials shall be of edible grade, and shall be clean, sound, wholesome, and free from evidence of insect infestation or any other objectionable foreign matter, and from any foreign flavor or odor.

3.2.1 Wheat flour. Wheat flour shall be of the commercial type known as cracker flour. It shall be bleached or unbleached and enriched or unenriched. If unenriched flour is used, equivalent enrichment shall be incorporated by the manufacturer at the time of production of the finished product.

3.2.2 Shortening. The shortening shall be hydrogenated cottonseed or peanut oil, or a combination of both, and shall have a stability of not less than 100 hours when tested by the active oxygen method (AOM) (see 4.6.3). The shortening shall contain antioxidants such as BHA, BHT, propyl gallate, or TBHQ singly or in combination to the maximum total amount permitted by FDA regulations.

3.2.3 Leavening-yeast. Leavening yeast shall be either compressed bakers or active dry yeast. The yeast shall be a culture of saccharomyces cerevisiae.

3.2.4 Yeast, dried, inactive. The dried yeast shall be either primary grown, properly debittered brewers, or torula yeast. It shall be mild in flavor, finely powdered, and shall comply with the National Formulary quality, except for vitamin potency. Sufficient quantities of fortifying agents shall be added to the yeast during processing to yield the following vitamin potencies:

	Micrograms per gram
Thiamin	Not less than 1200
Riboflavin	Not less than 600
Niacin	Not less than 1000
Pyridoxine	Not less than 600

3.2.5 Salt. Salt shall be white refined sodium chloride with or without anti-caking agents. Topping salt may be coarse or flaked. Iodized salt shall not be used.

3.2.6 Baking soda. Baking soda shall comply with the requirements of the Food Chemicals Codex. It shall be of a granulation commonly used in the cracker industry and shall be free flowing.

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3.2.7 Malt preparations. Malt preparation shall be derived from water extract or barley malt, dried or evaporated, with or without cereal adjunct, or from malted barley flavor and a fine dextrose hydrate, with or without cereal adjunct.

3.2.8 Calcium carbonate. Calcium carbonate shall comply with the requirements of the Food Chemicals Codex.

3.2.9 Protease enzyme. Protease enzyme shall be of high commercial grade and produced from either fungal or bacterial action.

3.2.10 Yeast food. Yeast food shall be of high commercial grade.

3.3 Formula. The crackers shall be prepared by the sponge-dough or straight dough method from ingredients in the following proportions:

Material	Pounds
Flour 2/	200
Shortening	28
Salt	1-1/2
Dried yeast, inactive	As required
Leavening yeast	As required
Baking soda	As required
Water	As required
Malt preparation, protease enzyme, yeast food (if necessary)	As required
Calcium carbonate ($CaCO_3$) 1/	

1/ In addition, class 2 (square) crackers shall contain added calcium carbonate at a rate to yield 2.5 grams per pound of finished crackers.

2/ Ground crackers derived from clean but broken or otherwise defective crackers from previous runs of this formulation may be added to each batch in an amount not to exceed 5 percent of the flour ingredient.

3.4 Topping salt. Topping salt shall be used on all classes of crackers at the rate of approximately 1 pound of salt per 100 pounds of baked crackers.

3.5 Finished product (see 6.3). The crackers shall be uniformly well baked but shall not be excessively brown in color. They shall be clean, reasonably tender and crisp, but not hard, shall show no indication of mold or foreign matter. Crackers shall have clean cut edges, and shall be free from dusting flour. The finished item shall be free from any foreign or objectionable taste and odor such as musty, rancid, burnt, stale, alkaline, soapy, or excessively salty.

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3.5.1 Classes.

3.5.1.1 Class 1 (rectangular). Class 1 cracker shall be of the following dimensions. Minimum weight of 4 crackers shall be not less than 1.5 ounces.

	Dimensions inches		
	Length	Width	Thickness
Maximum	3	2-3/8	5/16
Mnimum	2-15/16	2-1/4	1/4

☆ 3.5.1.2 Class 2 (square). Class 2 crackers shall be of the following dimensions:

Style		Dimensions - inches		
		Length	Width	Thickness (7 crackers)
A	Maximum	2-1/8	2-1/8	1-5/8
	Minimum	1-7/8	1-7/8	1-3/8

Number of crackers per pound shall not exceed 95. Crackers shall be in units of four crackers, arranged two by two, scored but not separated.

B	Maximum	4-1/4	4-1/4	1-5/8
	Minimum	3-3/4	3-3/4	1-3/8

Number of crackers per pound shall not exceed 24. Deep docker holes shall be placed regularly in the cracker, approximately 1/4 inch apart, with score lines of approximately 1/2 inch starting at the outside edge of the center row of docker pins in one dimension.

3.5.1.3 Class 3 (round). Class 3 cracker shall be of the following dimenions; inches:

	Diameter	Height (7 crackers)
Maximum	2-5/8	1-5/8
Minimum	2-3/8	1-3/8

Number of crackers per pound shall be from 61 to 65.

3.5.2 Allowable damage per container. The cracker shall be free from "checking" and breakage to the degree that when packaged as specified in 5.1, not more than ten percent of crackers shall be broken and not more than 20 percent shall be checked.

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* 3.5.3 Moisture content. Moisture content of the finished product shall not be greater than 4.0 percent or less than 1.5 percent on a unit basis. It shall not be greater than 3.5 percent or less than 2.0 percent on a lot average basis.

3.5.4 pH (hydrogen-ion activity). The pH of the finished cracker shall be not less than 6.5 nor more than 8.0.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act the Regulations Promulgated Thereunder.

3.7 Workmanship. The product shall be prepared, processed, and packaged under modern sanitary conditions and in accordance with good commercial practice. The product shall be processed in establishments meeting the requirements of MIL-STD-1105.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Plant qualification. The product shall be unacceptable if not produced and stored in plants which meet the requirements specified in 3.7.

4.4 First article inspection. When a first article is required, it shall be examined for the defects specified in table XIII. The presence of any defect shall be cause for rejection of the first article.

* 4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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* 4.5.1.1 Examination of ingredients and plastic bag components. Examinations shall be performed in accordance with 4.5.1.1.1, 4.5.1.1.2 and 4.5.1.1.3 to determine conformance to identity and condition requirement. Nonconformance to identity requirements, or one or more sample units not conforming to condition requirements shall indicate an unacceptable ingredient and use of such shall be cause for rejection of the involved finished product.

* 4.5.1.1.1 Identity of ingredients. Conformance to identity (kind, grade, quality, derivation, etc - as applicable) shall be ascertained by examination of pertinent labels, invoices or other valid documents.

* 4.5.1.1.2 Condition of ingredients. Each ingredient shall be examined organoleptically, as necessary, to determine conformance to 3.2 and condition requirements as may be contained in the applicable ingredient requirement paragraph. The sample unit shall be approximately 1 pound of each dry ingredient, 4 ounces for yeast and 4 fluid ounces of each liquid ingredient extracted from each of five primary containers. If a lot consist of less than five containers, a sample unit shall be extracted from each container in the lot.

* 4.5.1.1.3 Identity of plastic bag components. Conformance to identity (material, bonding, melt index, density, thickness, etc., as applicable) shall be ascertained by pertinent labels, invoices or other valid documents.

4.5.1.2 Testing of components. Testing of components described in this specification and changes in the testing of components described in subsidiary specifications shall be as shown in table I. Test requirements for the characteristics listed in the table shall be average requirements, except for seal strength, which shall meet both lot average and individual requirements. A test failure shall indicate an unacceptable component and use of such shall be cause for rejection of the involved end item.

*TABLE I. Testing of components

Component	Sample unit	Lot size expressed in	Inspection level	Characteristic	Results reported	Test ref. and rqmt. para
Shortening	1 pound composite 1/	Pounds	-	Stability	Pass or 2/ fail	3.2.2 4.6.3
Dried yeast, inactive	4 ounce 1/ composite	Pounds	-	Thiamine, Riboflavin, Niacin Pyridoxine	Nearest Microgram per gram	3.2.4 4.6.2
Box liner, glassine or greaseproof paper	1 liner	Liners	S-1	Basis wt. of paper Turpentine test	Nearest 1 pound Nearest second	5.1.1 4.6.4

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TABLE I. Testing of components (cont'd)

Component	Sample unit	Lot size expressed in	Inspection level	Characteristic	Results reported	Test ref. and rqmt. para
Carton liner, waxed glassine	1 liner	Liners	S-1	Basis wt. of paper Weight of wax Melting point of wax	Nearest 1 pound Nearest 1 pound Nearest 1°F	5.1.2 4.6.4
Carton liner, polyethylene bag	1 bag	Bags	S-1	Thickness 3/	Nearest 0.0001 inch	5.1.2
Carton or box overwrap	1 yard	Rolls	S-1	Basis wt. of paper Wt. of wax Melting point of wax	Nearest 0.1 pound Nearest 1 pound Nearest 1°F	5.1.1 5.1.2 4.6.4
Plastic bag	Bag	Bags	S-1	Seal strength 4/ Bag integrity 4/	Nearest 1/2 pound per inch Pass or fail	5.1.2.1 4.6.5 5.1.2.1.3 4.6.6

1/ The composite shall be derived from five primary containers.

2/ Report failure to nearest hour.

3/ In lieu of performing test for thickness, conformance to requirement may be ascertained by examination of pertinent labels, invoices or other valid documents.

4/ Separate samples can be used to test the bag manufacturer's seals and the end item closure seals. In addition, the bag integrity characteristic can be checked by performing tests before and after the end item is packaged, but must be checked on samples of each end item lot.

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4.5.2 Formulation certification. Formulation records shall be prepared and maintained by the contractor reflecting kind and quantity of ingredients as required by 3.3. Nonconformance as indicated by a review of the records shall be cause for rejection of the lot.

4.5.3 End item inspection.

* 4.5.3.1 End item examination. Examination shall be in accordance with tables IV through XIII and 4.5.3.1.1 and 4.5.3.1.2. Sampling plans for the tables of examination shall be in accordance with table II or III, as applicable. The acceptable quality levels (AQLs) shall be expressed as defects per hundred units.

*TABLE II. Sampling plans (other than Class 2, Style A crackers in alternative bulk package)

Table	Sample unit	Lot size expressed in	Inspection level	AQL	
				Major	Minor
IV	1 filled and wrapped carton or box	Cartons or boxes	S-3	0.65	2.5
VI	1 laminated bag	Bags	I	1.5	4.0
VII	1 filled carton or box	Cartons or boxes	S-3	2.5	4.0
VIII	1 filled box (class 1) 1 pound from a carton class 2 or 3)	Cartons, boxes or laminated bag	S-3	2.5	6.5
X	1 filled carton, or box	Cartons, boxes or laminated bag	S-3	2.5	10.0
XIII	1 filled carton or box (class 1 product) or 12 crackers from a carton (class 2 or 3 product)	Cartons, boxes or laminated bags	S-2	2.5	10.0

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*TABLE III. Sampling plans for class 2, style A crackers in alternate bulk package

Table	Sample unit	Lot size expressed in	Inspection level	AQL Major	AQL Minor
V	1 filled shipping container	Shipping containers	S-3	2.5	4.0
IX	1 pound	Pounds	S-3	2.5	---
XI	3 pounds	Pounds	S-3	2.5	10.0
XIII	1/2 pound	Pounds	S-3	2.5	10.0
XII	1 filled, marked, sealed and wrapped shipping container	Shipping containers	S-3	2.5	6.5

*TABLE IV. External examination of wrapped cartons or boxes (other than class 2, style A crackers in alternative bulk package)

Category		Defect
Major	Minor	
101		Overwrap(s) missing
102		Overwrap(s) not as specified
	201	Tear, hole, or open seal in one of two overwraps (for class 2 or 3 crackers)
103		Tear or hole or open seal in both overwraps (for class 2 or 3 crackers)
104		Tear, hole, or open seal in overwrap (for class 1 crackers)
	202	Overlap of each sheet less than 1/2 inch at seams and end closures (for class 2 or 3 crackers) 1/
	203	Longer seam of each overwrap not on opposite panels of carton
	204	Not clean
105		Labeling, if on carton or box, not legible through overwrap(s)

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TABLE IV. External examination of wrapped cartons or boxes (other than class 2, style A crackers in alternative bulk package) (cont'd)

Category		Defect
Major	Minor	
106		Nomenclature missing, incorrect, or illegible
	205	Other required label information missing, incorrect, or illegible

1/ Device used for measuring shall be calibrated in increments of 1/16 inch.

*TABLE V. Examination of internal packaging of shipping container (applicable to alternate bulk package (class 2, style A, crackers)

Major	Minor	
101		Polyethylene bag missing or not as specified
102		Corrugated parchment bag liner and/or dividers missing
103		Corrugated parchment liner and dividers missing or not positioned as specified
104		Tear, hole or open seam in polyethylene bag
	201	Polyethylene bag not thickness specified
105		Polyethylene bag and corrugated parchment not clean
	202	Crackers not packaged in uniform rows and layers
106		Corrugated box liner or top and bottom pads missing

*TABLE VI. Examination of plastic bag (other than class 2, style A crackers in alternative bulk package)

Category		Defect
Major	Minor	
151		Tear or hole or not heat sealed as specified
152		Color not as specified (see 5.1.2.1.1)
153		Dimensions not as specified

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*TABLE VI. Examination of plastic bag (other than class 2, style A crackers in alternative bulk package) (cont'd)

Category		Defect
Major	Minor	
	201	Tear notch missing or not located as specified
	202	Depth of notches not as specified
154		Not style or material specified
155		Not clean
156		Less than 3/16 inch between the inside edge of the tear notch and the inside edge of the seal
157		Required labeling missing, incorrect or illegible
	203	Closure seal extends below the tear notch on either side of the bag
158		One or more crushed or broken crackers 1/
159		Contents caught in closure seal
160		Contents crushed, adversely affecting serviceability

1/ See table X footnote 1/.

TABLE VII. Examination of internal packaging of cartons or boxes (other than class 2, style A crackers in alternative bulk package)

Category		Defect
Major	Minor	
101		Liner missing or not fabricated as specified
102		Liner not completely protecting product from interior surfaces of carton or box
	201	Bag liner stretched due to inadequate size of liner
103		Open end of bag liner not closed in the manner specified
104		Liner not clean

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TABLE VII. Examination of internal packaging of cartons or boxes (other than class 2, style A crackers in alternative bulk package) (cont'd)

Category		Defect
Major	Minor	
	202	Tips of corrugated liner not in contact with box (class 1 product)
	203	Removal of product from ends causes liner to tear (class 1 product)
	204	Class 2 or 3 crackers not packaged in uniform rows or layers

☆ TABLE VIII. Examination for net weight and count per pound (other than class 2, style A crackers in alternative bulk packages)

Category		Defect
Major	Minor	
101		Weight of four class 1 crackers less than 1.5 ounces 1/
102		Count per pound in cartons more than 95 for class 2, style A, or more than 24 for class 2, style B
103		Less than eight class 2, style A crackers or less than two class 2, style B crackers per plastic bag
104		Less than 12 crackers per box (class 1) 1/ 3/
	201	Net weight of class 2 or 3 crackers packaged in carton, more than 2 percent under labeled net weight 2/ 3/

1/ Make one determination per sample unit.

2/ Class 2 or 3 products in carton having a net weight of more than 3 pounds as indicated by labels or actual net weight shall not be acceptable.

3/ Report results to nearest 1/4 ounce. Lot shall be rejected if sample data indicate lot average net weight less than labeled net weight.

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*TABLE IX. Examination for count per pound (applicable to alternate bulk packaged class 2, style A, crackers)

Category	Defect
Major	
101	Count per pound more than 95

*TABLE X. Examination for condition 1/ and dimensions inches 2/ of finished product (other than class 2, style A crackers in alternative bulk package)

Category		Defect
Major	Minor	Class 1 - in boxes
	201	More than 1 broken cracker
101		One or more crushed cracker
	202	More than 2 checked crackers
	203	Size not as specified
		Class 2 or 3 - in cartons or bags
102		Broken or crushed, or broken and crushed crackers per pound five or more for class 3; or ten or more for class 2, style A; or three or more for class 2, style B
103		Checked crackers per pound ten or more for class 3; or nineteen or more for class 2, style A; or five or more for class 2, style B
	204	Size not as specified
		Class 2 - in cartons or class 3 - in cartons
104		Height of 7 crackers more than maximum requirement
	205	Height of 7 crackers less than minimum requirement

1/ A broken cracker is one than is broken into not more than three distinct pieces or has more than 1/8 but less than 1/4 of its area missing. Separation of cracker units along the score lines of class 2, style A crackers shall not be considered breakage. Corner breakage of the class 2, style B crackers which leaves at least 7/8 of the unit intact and breaks along the score lines shall not be considered breakage. A crushed cracker is one than is broken or shattered into more than three pieces or has 1/4 or more of area missing. A checked cracker is a whole cracker that has fracture lines appearing on the surface.

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2/ One determination per sample unit shall be made for dimensions. Device used for measuring shall be calibrated in increments of 1/16 inch.

*TABLE XI. Examination for condition 1/ and dimensions (inches) 2/ of finished product (applicable to alternate bulk packaged class 2, style A, crackers)

Category		Defect
Major	Minor	
101		Ten or more broken or crushed, or broken and crushed crackers per pound
102		Nineteen or more checked crackers per pound
	201	Size not as specified
103		Height of 7 crackers more than maximum requirement
	202	Height of 7 crackers less than minimum requirement

1/ A broken cracker is one that is broken into not more than three distinct pieces or has more than 1/8 but less than 1/4 of its area missing. Separation of cracker units along the score lines shall not be considered breakage. A crushed cracker is one that is broken or shattered into more than three pieces or has 1/4 or more of its area missing. A checked cracker is a whole cracker that has fracture lines appearing on the surface.

2/ One determination per sample unit shall be made for dimensions. Device used for measuring shall be calibrated in increments of 1/16 inch.

*TABLE XII. Examination of shipping containers (applicable to alternate bulk packaged class 2, style A, crackers)

Category		Defect
Major	Minor	
101		Not constructed as specified
102		Not closed as specified
103		Marking missing, incorrect or illegible
	201	Net weight more than 2 percent under labeled net weight
	202	Not wrapped as specified

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TABLE XIII. Examination for product quality 1/

Category		Defect
Major	Minor	
101		Not uniformly well baked
102		Excessively brown in color
103		Topping salt omitted
104		Not tender and crisp 2/
105		Hard 2/
	201	Edges not clean cut
	202	Not free from dusting flour
106		Objectionable taste or odor (e.g., musty, rancid, burnt, stale, alkaline, soapy, excessively salty) 2/

1/ Presence of foreign material or mold (e.g., dirt, filth, glass, paint, metal, wood, insects etc.) shall be cause for rejection of the lot.

2/ Examine one cracker per sample unit.

4.5.3.1.1 Examination of vacuum packaging of class 2 crackers in bags. Examination of vacuum gage readings shall be made to determine compliance with the vacuum limits of 5.1.2.1.3. Nonconformance shall be cause for rejection of the involved quantity of product.

* 4.5.3.1.2 Vacuum examination of class 2 crackers packaged in bags. The filled and sealed bags shall be examined not less than 48 hours after the vacuum packaging specified in 5.1.2.1.3. The unit pack shall continue to exhibit a tight adherence of the bag walls to the surface contours of the contents along the seams when a force is applied away from the contents at the center of each side seam. The force is applied by holding each side seam between thumb and forefinger of each hand, simultaneously exerting a slight pull with both hands. The sample unit shall be one filled and sealed bag. The lot size shall be expressed in units of bags. The inspection level shall be S-4. Any nonconformance shall be cause for rejection of the lot; however, withdrawal or repackaging (within 24 hours of inspection) is permitted.

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* 4.5.3.2 End item testing. The finished product shall be tested for moisture and pH requirements in 3.5.3 in accordance with 4.6.1. The sample unit shall be one filled and sealed primary container, except for class 2, style A crackers in alternate shipping containers, for which the sample unit shall be 1/4 pound of product. Sample size shall be the number of sample units indicated by inspection level S-2. The lot size for purpose of determining sample size shall be expressed in pounds. A moisture sample with more than 4.0 percent or less than 1.5 percent, or an average result of less than 2.0 percent or more than 3.5 percent shall be cause for rejection of the lot. An average pH result not within the specified limits shall be cause for rejection of the lot.

4.5.4 Packing inspection. Examination of the filled and closed shipping containers shall be in accordance with the examination criteria of the appendix to PPP-B-636. In addition, the following defects shall be included in the table of examination; Major - Marking; missing, incorrect, or illegible; Minor: count not as specified (see 5.2.2); cushioning material not placed as specified (see 5.2.2). Shipping containers containing boxes or cartons shall not be acceptable if markings indicate a total net weight of more than 45 pounds.

4.6 Methods of inspection.

4.6.1 Moisture and pH tests. Moisture and pH determination shall be made in accordance with the following methods of Official Methods of Analysis of the Association of Analytical Chemists, Chapter: Cereal Foods, section: Wheat Flour:

Test	
Moisture	Total Solids (Moisture, Indirect Method)
pH	Hydrogen-Ion Concentration (Electrometric Method)

4.6.2 Vitamin potency test. Vitamin potency (thiamine, riboflavin, niacin and pyridoxine) of yeast shall be made by Methods of Vitamin Assay of the Association of Vitamin Chemists.

4.6.3 Shortening stability test. Shortening shall be tested for stability (AOM) in accordance with Official and Tentative Methods of the American Oil Chemists Society, Chapter: Commercial Fats and Oils, method Cd 12-57.

4.6.4 Paper tests. Paper shall be tested in accordance with the following methods of the FED-STD-101.

Basis weight	5022
Turpentine test	3017
Melting point of paraffin wax	4017
Paraffin in paper	6018

4.6.5 Seal strength test. Seal strength shall be determined in accordance with the procedures specified in Method A or B of ASTM D 882, except that testing speed may be 10 or 12 inches per minute. Samples shall be taken by taking three adjacent specimens 1/2 or 1 inch wide from each edge of the bag.

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* 4.6.6 Test for determining bag integrity and air-tightness of heat seals. Bag integrity and air-tightness shall be determined by placing crackers, fiberboard or similar rigid material approximately 4 inches by 4 inches by 3/8 inch into the bag and sealing under a vacuum of not less than 15 inches of mercury. Not less than 48 hours after vacuum sealing, the package shall be examined for vacuum loss in accordance with 4.5.3.1.2. Any nonconformance shall be cause for rejection of the involved quantity of product.

5. PACKAGING

5.1 Preservation. Preservation shall be level A in accordance with 5.1.1 and 5.1.2, as applicable.

5.1.1 Class 1 crackers. Twelve crackers shall be unit packed in a box conforming to variety 1, style II, type E, class a of PPP-B-566. Alternatively, the top may be style II and bottom style I, type A, class a. The board shall be of a bending grade manila, or solid sulphate in accordance with PPP-B-566. The box shall be lined with a single-faced corrugated liner made from two sheets of glassine or greaseproof paper. Each sheet shall have a nominal basis weight of not less than 25 pounds per ream (24 by 36-500) and have a minimum average turpentine test of 300 seconds (see 4.6.4). The liner, formed from a single piece of material, shall be inserted into the box so as to cover all inside surfaces and allow for removal of the contents from either end of the box without tearing the liner. The corrugated side of the liner shall be against the box. The filled and closed box shall then be overwrapped with two sheets of waxed sulphite or waxed bleached kraft paper. Each sheet shall have a nominal basis weight of not less than 28 pounds per ream (24 by 36-500), and shall be wet waxed with 14 pounds per ream (24 by 36-500) of fully refined paraffin wax having a melting point of not less than 130°F (see 4.6.4). The wrappers shall be fully heat sealed and shall not impart odor or flavor to the product.

* 5.1.2 Classes 2 and 3 crackers for ration use. Not more than 3 pounds of the product shall be unit packed, in uniform rows and layers, in a good commercial carton. The carton shall be lined with a waxed, unbleached, glassine paper having a nominal basis weight of not less than 25 pounds per ream (24 by 36-500). The sheet shall be waxed with 3 pounds per ream (24 by 36-500) of a fully refined paraffin wax having a melting point of not less than 130°F. The wax shall be odorless and tasteless. The liner shall overlap itself sufficiently to insure adequate protection of the contents from the board. Alternatively, the carton shall be lined with a bag constructed of polyethylene film having a nominal thickness of 0.0015 inch when tested in accordance with the requirements of L-P-378. The bag liner shall be of sufficient size to inclose the crackers without stretching. The bag shall be of sufficient length so that the open end may be closed by pressing opposite sides together and folding the open end with three or more folds to form a moisture proof seal, alternatively, the polyethylene inner liner may be closed with a twist tie closure forming a moisture proof seal. The film shall conform to the requirements of the Federal Food, Drug, and Cosmetic Act, for contact with food. The filled and closed carton shall then be overwrapped with two sheets of waxed paper, two sheets of cellophane, or a combination of one sheet of waxed paper and one sheet of

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cellophane. The waxed paper overwraps shall be made from a good quality sulphite or kraft paper having a nominal basis weight of 23 pounds per ream (24 by 36-500), and shall be wet waxed with 14 pounds per ream (24 by 36-500) of a fully refined paraffin wax having a melting point of not less than 130°F. The cellophane overwraps shall be type IV, class 2, having a nominal thickness of 0.0009-inch in accordance with L-C-110. Both sheets shall be large enough to effect an overlap at the seam and end closures of not less than 1/2 inch. Each wrap shall have a continuous heat seal made along the full length of the overlaps. The longer seams of the two sheets shall be located on opposite panels of the carton. Alternatively, approximately 18 pounds of class 2, style A, crackers may be unit packed in a 0.002 inch thick polyethylene bag which is lined with a U-shaped single faced corrugated parchment sheet running the length of the shipping case inside the bag from the top side down across the bottom and up the opposite side to the top. The bag shall be secured with a twist tie or comparable closing device at the top. Crackers shall be arranged in two layers of three rows each, with each layer and row separated by a corrugated parchment pad. Each bag of class 2, style A crackers shall be packed in a 275 pound test, C-flute, RSC-L container, approximately 16 by 12-7/8 by 8-5/16 inches, with top and bottom corrugated pads. The container shall be glue sealed and shall be double shrink wrapped with not less than 0.0015 inch thick polyethylene.

* 5.1.2.1 Class 2 crackers in plastic bag. When specified (see 6.1), eight class 2, style A crackers (two units of four crackers each) or two class 2, style B crackers shall be unit packed in bags constructed from the material specified in 5.1.2.1.1. The bags shall be constructed in accordance with 5.1.2.1.2 and shall be filled and sealed (closed) as specified in 5.1.2.1.3.

* 5.1.2.1.1 Bag material. The material shall be fabricated by melt extrusion of 0.003 inch thick (± 0.0003 inch) ionomer to 0.00035 inch thick (± 0.000035 inch) aluminum foil, and the bonding of 0.0005 inch thick (± 0.00005 inch) polyethylene terephthalate to the opposite side of primed aluminum foil by melt extrusion with 10 pound ream low density polyethylene. The ionomer shall be the zinc neutralized copolymer of ethylene and methacrylic acid, having a melt index of 5g/10 minutes in accordance with ASTM D 1238 and a density of 0.936 g/cc in accordance with ASTM D 1505. The exterior (polyester) surface of the bag shall be colored overall with a color in the range from 34079, or 34087, or 24052 through 24087, or 10045 or 30045 through 30118 (except that 30109 may not be used) of FED-STD-595. The material shall be suitably formulated for food packaging and shall not impart an odor or flavor to the product being packaged. The material shall show no evidence of delamination or degradation when heat sealed or fabricated into bags.

* 5.1.2.1.2 Bag construction. The bag shall be flat style bag having inside dimensions of 5-1/2 by 6 inches (± 1/8 inch), with the 5-1/2 inch dimension being the opening of the bag. The empty bag shall be made by heat sealing all three edges with 3/8 inch (± 1/8 inch) wide seals. The heat seals shall be made in a manner that will assure hermetic seals and preclude leakage of air into the package through the seals, seal junctions or through defects created when the bags are filled and

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sealed as specified in 5.1.2.1.3. Integrity of the bag and air tightness of the seals shall be tested in accordance with 4.6.6. The side and bottom heat seals shall have an average strength of not less than 6 pounds per inch and no individual specimen shall have a strength less than 5 pounds per inch (see 4.6.5). A V-shaped or C-shaped (half round) tear notch at least 1/32 inch deep, located 1 to 1-1/4 inch from the top edge of the bag, shall be made in one or both side seals. The distance between the inside edge of the tear notch and the inside edge of the seal shall be at least 3/16 inch. One side of the open end of the bag may be provided with a 1/4 ± 1/16 inch extended or foldover lip to facilitate opening and filling; if so, tear notch location shall be measured from the top of the bag.

* 5.1.2.1.3 Filling and sealing cracker bags. The crackers shall be filled into the bags in such a manner as to avoid puncturing the material. The filled bag shall be closed under a vacuum of not less than 15 or more than 20 inches of mercury (see 4.5.3.1.2) with a 3/8 ± 1/8 inch wide heat seal. If thermal impulse closure seals are used, a seal width of 1/8 inch will be acceptable. The closure seal shall not extend below the tear notch on either side of the bag. The closure seal shall have an average strength of not less than 6 pounds per inch and no individual specimen shall have a strength less than 5 pounds per inch (see 4.6.5) and shall meet the requirements of the bag integrity test (see 4.6.6).

5.2 Packing. Packing shall be level C.

5.2.1 Level C, for shipment to ration assembler.

5.2.1.1 Cartons and boxes. Not more than 45 pounds of the product, preserved as specified in 5.1, shall be packed in a shipping container in accordance with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.2.1.2 Laminated plastic bags. Ninety-six unit packs shall be packed flat in a snug-fitting fiberboard box constructed and closed in accordance with style RSC-L, type CF, class domestic, variety SW, grade 175 of PPP-B-636. Cushioning material not less than 1/8 inch thick in accordance with type II, class B of PPP-C-843 or cushioning material not less than 1/16 inch thick in accordance with PPP-C-1797 shall be placed between each of the layers and at the bottom and the top of the box to minimize movement of individual packages.

5.3 Marking.

5.3.1 Unit packs. Unit packs shall have the following information permanently imprinted on the overwrap or on the carton if the printing is legible through the overwrap:

NAME OF PRODUCT (in letters larger than any others used for the label)
Net weight 1/
Ingredient statement 1/
Name and address of manufacturer
Any other information required by the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder 2/

1/ Optional for class 2 and class 3 crackers.

2/ Not applicable to class 2 and class 3 crackers.

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5.3.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The crackers are intended for use by the Department of Defense as a component of operational rations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Class of product required (see 1.2).
(c) When a first article is required (see 3.1, 4.4 and 6.4).
(d) When class 2 crackers are required to be packaged in laminated plastic bags (see 5.1.2.1).

6.3 Cracker size, shape, height and weight. To aid in maintaining correct size and shape of crackers, as well as height, and weight (see 3.5), and hold blistering and "checking" to none or to-a minimum, the following is listed for guidance:

(a) Increase docker pins to 40 or more per cup.
(b) Enclose the cooling system.
(c) Control cooling conditions from the oven to the packer, to inhibit "checking" and shattering.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample. The first article should consist of three pounds of finished product. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

Project No. 8920-0427

Review activities:

Army - MD
Navy - MC, MS
DP - SS

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